

Build Cost Calculator: Trade Expansion and Beyond Plan

Version 2 | 1 October 2026 | Written against js/calculator-page.js (2,905 lines), calculator.html, sw.js, the QA harnesses and AGENTS.md from your repository

What changed from v1. v1 was written without your code. v2 is built on it. Reading the source overturned three things I told you: formwork already exists as a derived item; there is no labour-hours layer anywhere in the engine; and the currency selector never converts a rate. Section 1 lists everything found. Section 7 is new: the instructions for making this calculator genuinely ahead of anything in the market.

1. What your code actually does (audit findings)

The engine is well built. computeFor(st) is a pure function over one settings object, history and estimates recall exactly, and the bill, site costs, contingency, escalation, cash plan and quote printing are all there. The findings below are the places the code stands between you and the JIC-aligned, best-in-market goal.

ID	Finding	Evidence in the code	Consequence	Fixed in
A1	Currency never converts. Choosing JMD only changes the symbol. Model rates are single numbers in one implied currency, and every location pack has index 1.	CURRENCY map holds symbols only; computeFor has no exchange step; the JM pack note says rates are used as entered. Model slab labour is 85 per m3, which only makes sense as USD.	A Jamaican user who picks JMD sees J\$85 per m3 labour. Totals look plausible and are wrong by roughly two orders of magnitude.	Phase 1
A2	One rate pair per item. Each WORK entry has a single {mat, lab}. There are no variants.	WORK table, lines 38 to 150.	JIC prices by soil type, block size, pipe diameter, thickness and height band. None of that can be expressed today.	Phase 1
A3	Unit mismatch. The engine speaks m2, m3, m, t, L, each, point, square. JIC speaks yd2, ft2, ft run, in., lb, dozen.	UNIT_CONV covers only m2, m3, m, L.	Rates keyed in JIC units need a conversion layer or contractors will re-key everything.	Phase 1
A4	Rate sheets are not rate books. A saved sheet holds one item's material, labour, equipment and piece price.	applySheetById fills only the current item's fields; importSheets validates rateMat and rateLab only.	A full JIC schedule (hundreds of lines) cannot be saved or switched as one object.	Phase 1
A5	No labour-hours or productivity layer. Labour is a flat money rate per unit.	No productivity or crew logic in the file; your own estimating-depth spec lists it as an open gap.	No crew-days, no duration, no schedule handoff, no worker-pay sanity check.	Phase 9
A6	Formwork is derived, one rate, not pickable. It prices contact area at one rate per m2 and appears only as a companion line.	WORK.formwork is bill-only; W7 derived-formwork box.	JIC's element-by-element formwork rates cannot be applied. (This corrects my v1.)	Phase 2
A7	Release chain is heavier than it looks. calculator-page.js is not in the bundles. It loads straight from calculator.html with ?v=22. sw.js has a shell cache (v347) that verify:sw forces you to bump.	calculator.html script tags; sw.js SHELL list and CACHE constant; build.js has no calculator-page entry.	Every calculator change needs: bump ?v, bump the sw.js cache, run node build.js if CSS changed, run npm run verify. Skip one and users keep stale code.	All phases
A8	Two test baselines will break on every new item. qa-calculator-page has 164 gates; qa-calc-trade-coverage walks all 33 items through the real UI (68 gates) and is registered in CI tier T4.	Header of tools/qa-calculator-page.cjs; sw.js v347 note ('picker now 33 items, FM5 re-baselined').	A new item with no harness update turns CI red. This is also why the plan makes tests part of the item, not an afterthought.	All phases

ID	Finding	Evidence in the code	Consequence	Fixed in
A9	Server never reads the estimate. Workspace sync stores whatever JSON the page sends and merges by section timestamps.	src/cloud/calc-workspace.js: 'the server never interprets the sections'.	Good news: new state fields (variant, rate book id, and so on) sync across devices with zero server work.	n/a
A10	Owner decision on record: no AI on the calculator page.	sw.js v325 note.	Every 'smart' feature in section 7 is a deterministic algorithm that runs on-device. None calls a model.	Section 7

2. Rules that apply to every phase

- **Additive only.** Never rename or remove a WORK key. Saved estimates, history and bill lines recall by key, and the family filter is visual only.
- **The engine stays pure.** New math goes in functions that take a settings object and read no DOM. This keeps tests simple and makes the uncertainty and cutting-stock features in section 7 possible.
- **Offline-first is sacred** (AGENTS.md rule 3). No live rate feeds. Rate books and updates arrive as imported JSON.
- **Release checklist, every change:** bump ?v on calculator.html assets; bump the sw.js cache; node build.js if CSS changed; npm run verify; update tools/qa-calculator-page.cjs, tools/qa-calc-trade-coverage.cjs and docs/CI-TEST-COVERAGE.md in the same change.
- **Guardrails from AGENTS.md:** no inline scripts (CSP hashes), no emoji, sprite icons only, colours from theme tokens so light and dark both hold, Conventional Commits with no AI-credit footer.
- **No test, no ship.** Every item and every feature ships with a hand-calculated expected result. A difference of more than 1 currency unit rejects it.
- **Every rate carries a source tag:** JIC, Independent, Model or User. A rate with no tag does not reach a client document.
- **One phase at a time.** A phase starts only when the previous gate passes in full. A failed check is fixed, not waived.

3. Gap register: trades and items to add

Tags: **JIC** = confirmed in the 2025-2027 extract you sent. **Verify** = seen on the independent contractors' sheet or in market practice; confirm against the official IMAJ sheet. **Added** = my addition from industry practice. Where an item says *variant*, it is one work item with a selector (see Phase 1), not many near-duplicates.

A. Confirmed in your JIC extract

ID	Trade / item	What to add	Unit	Tag
G1	Formwork by element (variant)	Columns; beams; belt and stiffener; suspended slab (floor); suspended slab (stairs); walls; circular; sides and deck of manholes; new formwork (reuse). Makes the derived formwork line JIC-correct.	ft2	JIC
G2	Scaffolding (variant)	To 10 ft unbraced; 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80 ft tied to building	ft2	JIC
G3	Welding and metal cutting (variant)	Cut and sand smooth by thickness: torch 1/8, 1/4, 3/8, 1/2, 5/8, 3/4, 1 in; CPSAW 1/16 and 1/8 in	in.	JIC
G4	Steelwork (variant)	Rebar by size 3/8, 1/2, 5/8, 3/4, 1 in (per lb); stirrups 1/4, 3/8, 3/8 large girth (per dozen); fabric mesh (per yd2). Adds a by-weight mode beside your 85 kg per m3 norm.	lb / doz / yd2	JIC
G5	Masonry (variants)	Block 6 or 8 in; fill all or alternate pockets; ground to first floor versus first floor and above; manholes, drains, gully basins	yd2	JIC
G6	Tiling and terrazzo (variant)	Terrazzo lay and grout; first cut; final cut and polish; upper-floor extra; treads 10 and 11-12 in; risers 6-8 in; marble floors and walls	yd2 / ft run	JIC
G7	Excavation (variant)	Compacted earth, marl, sand, stiff clay (to 5 ft, 5-10 ft), asphaltic concrete; rock with no compressor, with compressor incl. labourers, labourers only	yd3	JIC
G8	Painting (variant)	Walls, ceilings, pebble dash (one and two coats); skirting 3-4 in; cutting to line	yd2 / ft run	JIC

ID	Trade / item	What to add	Unit	Tag
G9	Plumbing pipe (variant)	Diameters 1.25, 1.5, 2, 3, 4, 6 in; push-fit 200, 250, 300 mm (extract shows feet; confirm inches). Replaces the single per-metre rate.	ft run	JIC
G10	Electrical conduit (variant)	1/2, 3/4, 1, 1-1/4, 1-1/2, 2 in, plus the second 1 to 1-1/2 in set whose qualifier is missing from the extract	ft run	JIC
G11	Skirting matrix (variant)	1x3, 1x4, 1x6 in across bevelled top, bevelled mahogany, moulded top (WPP), moulded mahogany	ft run	JIC

B. Probably on the full JIC or independent sheets

ID	Trade / item	What to add	Unit	Tag
B1	Concrete labour	Rod and settle premix; fill and ram flooring; screeding by area	yd2	Verify
B2	Roofing labour	Sheeting, timber roof framing, ridge cap, flashing	per sheet	Verify
B3	Demolition	Break out block walls and slabs; tile and screed removal	by unit	Verify
B4	Waterproofing and sealing	Concrete roof slab waterproofing, sealants	yd2	Verify
B5	Windows and doors (variant)	Louvre, sliding, burglar-barred; fit versus supply-and-fit	each	Verify
B6	Plaster and render (variant)	Wall, ceiling, pebble dash as separate items	yd2	Verify
B7	Sheds and carports	Light steel or timber framed	each	Verify

C. Beyond the sheets

ID	Trade / item	What to add	Unit	Tag
C1	Stairs	Concrete and timber stairs; handrails and balustrades	per tread / m	Added
C2	Retaining and boundary walls	Retaining wall with footing and steel; boundary wall; gates	m	Added
C3	Drainage and external works	Storm drains, culverts, soakaways, French drain, pavers	m / each	Added
C4	Roof structure and tie-downs	Trusses, purlins, hurricane straps and tie-downs	m2 / each	Added
C5	Insulation and non-tile floors	Roof insulation, ceiling boards, timber, vinyl, laminate	m2	Added
C6	Electrical fit-out	Lighting points, switches and sockets, ceiling fans, earthing, generator transfer switch, solar PV	each	Added
C7	Mechanical	Split AC install, water tank and pump, solar water heater	each	Added
C8	Security and fit-out	Burglar bars, CCTV, countertops, wardrobes, bathroom accessories	m / each	Added
C9	Landscaping and termite treatment	Topsoil and sod; soil pre-treatment	m2	Added

Adding one item in your codebase touches six places: the WORK entry (with q(), rate, matDesc, waste or piece settings); the optgroup option in calculator.html; WORK_FAMILY; INSTANCE_KINDS if it is measured by wall, area, pour or run; COMPANIONS if it implies other work; and the two test harnesses. A new family (Temporary and metal works, Demolition and external works) also needs an entry in FAMILIES and the family select.

4. Foundation engine upgrades (Phase 1, before any new trade)

ID	Upgrade	Design (fits the existing code)
E1	Currency and FX layer (fixes A1)	Store model rates in a declared base currency. Add an editable, date-stamped FX table with a stale warning, and convert at the one point computeFor reads rates. Jamaica ships a native JMD rate book so FX is only the fallback. Existing saved estimates carry a currency tag so they recall to the same total they showed. Do not guess an exchange rate: the user or the rate book supplies it, with an as-of date.

ID	Upgrade	Design (fits the existing code)
E2	Variant engine (fixes A2)	Add an optional variants map to a WORK entry: { id, label, rate: {mat, lab}, unit }. One new calc-variant select appears when a variant set exists. A new variant field rides readState, applyState, history, BOQ lines, CSV and print. Missing field on a legacy save defaults to the first variant, so old recalls reproduce the same total.
E3	Unit adapter (fixes A3)	Extend UNIT_CONV with yd2, ft2, ft run, in., lb and dozen as display and rate-entry units. The math stays metric underneath, as it does today. Each work item declares its JIC unit; the rate field label follows it.
E4	Rate book v2 (fixes A4)	A whole-book object: { id, name, source, effective_from, effective_to, tier, currency, rates: { workKey: { variantId: {mat, lab, unit} } }, checksum }. Imports and exports as JSON through the existing import pattern; validation rejects unknown keys and negative rates. JIC 2025-2027 is the first book, with an expiry banner after the end date.
E5	Golden-case harness	One JSON file of hand-calculated cases (inputs, rate book, expected quantity, expected total) that both QA harnesses read. A new item is invalid without a case.
E6	Labour mode	A pricing mode that prices labour only at the book rate (JIC rates are labour piece rates). Materials come from the user's own rates or are omitted.

Licensing check before shipping JIC rates. The official sheet is a paid IMAJ publication. Putting its figures into a product other people use may need IMAJ's written permission. This is not legal advice, so confirm with them. A safe fallback: ship the JIC structure (items, units, variants) with empty rates and let each user import the rate book they bought. The rate-book design in E4 supports both routes, so decide before Phase 1 and the schema does not change.

5. Phases and exit gates

Phase	Scope	Exit gate (every line must pass)
0. Baseline	Obtain the official IMAJ sheet. Settle licensing. Run npm run verify and both harnesses green. Write golden cases (E5) for all 33 existing items.	33 of 33 items have a golden case and pass. verify and CI green before any change. Licensing route written down.
1. Foundations	E1 currency and FX, E2 variants, E3 units, E4 rate book v2, E5 golden harness, E6 labour mode. No new trades.	Old estimates recall to identical totals. A JMD estimate prices from the JMD rate book, not relabelled USD. Excavation converted to variants with unchanged results for old saves. ?v and sw cache bumped. CI green.
2. Concrete chain	G1 formwork, G4 steel, G7 excavation, G5 masonry, B1 concrete labour.	A footing-to-slab estimate prices end to end with no manual lines. Every variant has a passing golden case, metric and imperial.
3. Access and metal	G2 scaffolding, G3 welding, stirrups and fabric. New family: Temporary and metal works.	Every height band and thickness passes. New family shows in the filter and hides no old item.
4. Finishes	G6 tiling and terrazzo, G8 painting, G11 skirting, B6 plaster, B5 windows and doors.	Rates and units match the official sheet line for line. Terrazzo case covers lay, cuts, polish and the upper-floor extra together.
5. Services	G9 plumbing, G10 conduit, C6, C7.	Diameter changes the rate in every case. The unresolved electrical qualifier is answered from the official sheet, not guessed.
6. Remaining trades	B2, B3, B4, B7, C1 to C5, C8, C9, one cluster at a time.	Each cluster meets the definition of done (section 6) before the next starts.
7. Trust	X1 estimate linter, X2 uncertainty ranges, X3 provenance and verifiable share link.	Section 7 gates for X1 to X3 pass. Linter has zero false positives on the 33 golden cases.
8. Speed	X5 quick-add parser, X6 measure-on-plan, X7 assemblies.	Parser reproduces 30 written test phrases exactly. Plan measurement agrees with a known drawing to within 1 percent.
9. Beyond estimating	E-productivity layer, X8 estimate-to-project handoff, X9 procurement and cutting-stock.	A bill pushes into a My MaNaGeR project as budget lines and WBS tasks and the project's own checks show no new warnings.
10. Intelligence	X4 learn-from-actuals, X10 Jamaica intelligence, X11 carbon (optional).	Calibration never changes a rate without the user accepting it. Every factor has a cited source and date.
11. Resilience and release	X12 self-updating offline install, X13 accessibility and language, final regression.	Calculator works from a cold offline start on a fresh device. WCAG 2.2 AA checks pass. Guide and tour match every new field.

6. Definition of done for every work item

1. In the right family and picker group; old keys untouched.
2. Dimension labels correct in metric and imperial.
3. Variant selector wherever the JIC sheet splits the rate.
4. Rate comes from the rate book, is editable, and carries a source tag.
5. Unit shown in the JIC form and converted correctly.
6. Waste default set where the work has waste.
7. Productivity figure recorded (once the labour layer exists in Phase 9; a blank is allowed before).
8. Companion suggestions linked where one applies (formwork with rebar and slab, scaffolding with high walls).
9. Save, recall, history, bill line and named estimate round-trip identically; workspace sync carries it.
10. Print, CSV and the quote layout show the line correctly.
11. Light and dark themes both render; no inline scripts; no emoji.
12. Golden case passes in both harnesses, and docs/CI-TEST-COVERAGE.md is updated.

7. Making it way beyond its time

These are the features I did not find in the calculators I looked at (a proper market survey is still to do, so treat 'ahead of the market' as a hypothesis to test). They are chosen because your architecture (pure engine, on-device data, one settings object) makes each one buildable without a server or a model. Each has a gate. Each one runs offline.

Pillar 1: an estimate you can defend

ID	Feature	What it does and why it is ahead	How to build it here	Gate
X1	Estimate linter	A second QS in your pocket. After Calculate, rule checks flag what is probably missing or wrong: rebar with no concrete, slab with no formwork or excavation, walls with no footing, no contingency on a multi-month build, bill total outside the benchmark band for floor area, a rate far from its own history. Each flag explains itself and offers a one-tap fix.	A list of pure rule functions over the bill (BOQ array) and settings. Reuse the benchmark note already in the Jamaica pack. Same one-tap-add pattern as COMPANIONS.	Every rule has a failing and a passing test. No flag on any golden case unless intended.
X2	Ranges, not single numbers	Shows P10, P50 and P90 totals, and a driver chart of which lines move the answer most. The page already says 'about 15 percent under to 30 percent over'. This replaces that sentence with a computed range for this specific bill.	Run computeFor 2,000 times with rates and quantities drawn from triangular distributions (low, model, high per tag: JIC tight, Model wide). Pure engine makes it fast. Pick the AACE class band from how much of the bill is measured versus assumed.	Same seed gives the same range. P50 within 1 percent of the single-point total.
X3	Provenance and verifiable share link	Every figure shows where it came from (JIC line number, rate book and date, or your own). The quote can carry a share link that reopens the exact estimate on any device with no account, and a checksum so a client can confirm nothing was altered.	Compress the settings object (browser CompressionStream where present), base64url into the URL fragment so it never reaches a server. Add a hash of the canonical JSON. Optional QR on the printed sheet from a small vendored library (CSP: script-src self).	Round-trip: open the link on a clean browser, totals identical. A one-character edit to the link fails the checksum.
X4	Learn from your actuals	The user enters what jobs really cost. The calculator computes a personal correction per trade and offers it as a suggested rate. Over time the model becomes the contractor's own, on-device, no cloud.	A local actuals ledger, then a shrinkage estimate that pulls the model rate toward the user's own average, weighted by how many jobs exist. Never auto-applied; always shown with before and after.	With 0 jobs, no change. With known inputs, the suggestion matches a hand calculation.

Pillar 2: speed of capture

ID	Feature	What it does and why it is ahead	How to build it here	Gate
X5	Quick-add by typing or saying it	Type 'slab 10 x 8 x 150mm' or 'block wall 12 by 2.4, two windows' and the form fills itself. This is a deterministic grammar, not AI, so it respects your no-AI decision and works offline.	A small tokenizer and rule table mapping words to WORK keys and dimensions, writing into the same readState shape. The app already has a voice module (mmgr-voice.js) that can feed it.	30 written phrases produce the exact expected state. Unknown words ask a question; they never guess.
X6	Measure on the plan	Upload a plan image or photo, click two points of known length to set the scale, then click out lengths and areas. Results drop into the element rows as walls, pours and runs. Removes the retyping step where most estimating errors start.	Canvas overlay, scale calibration, polyline length and shoelace area, all on-device. Images first; PDF only if a reader is vendored into the repo.	A known 10 m x 6 m drawing measures within 1 percent.
X7	Assemblies	One tap adds a whole element: a column (excavate, footing, steel, formwork, concrete), a bathroom, a boundary wall, a two-bedroom shell. Parametric: change one dimension and every line updates.	A recipe table of WORK keys and quantity formulas over shared inputs, adding bill lines through the same path as COMPANIONS.	Each recipe equals the sum of its hand-priced parts.

Pillar 3: beyond estimating

ID	Feature	What it does and why it is ahead	How to build it here	Gate
X8	Estimate to schedule to budget to EVM	The bill becomes budget lines and WBS tasks in My MaNaGeR, with durations from labour hours and crew size, and the cash plan pre-loaded. I know of no calculator that connects a quote to a live earned-value project in one product, though I have not surveyed the market.	Add the labour layer (productivity per item and crew mix), then a Send-to-project action using the same project state the app already merges and validates. Your schedule engine does the dates.	A pushed bill produces tasks whose durations match crew-days. The project's own schedule checks report no new warnings.
X9	Procurement and waste optimiser	Turns the bill into a supplier-ready order list and cuts waste: a 1D cutting-stock optimiser for rebar against stock lengths, tile and board layout counts, block and bag totals rounded to what actually ships.	First-fit-decreasing for bar cutting with user-set stock lengths; existing piece-count engine for tiles, blocks and sheets.	Optimised waste is never worse than the plain percentage method on 10 test cut lists.
X11	Embodied carbon beside cost (optional)	Shows kg CO2e next to dollars for concrete, steel, block and timber, and suggests a lower-carbon alternative with its cost change.	An on-device factor table with a cited source and date (for example the University of Bath ICE database; confirm the current version before use).	Every factor has a source, version and date. No factor without all three.

Pillar 4: Jamaica-first and built to last

ID	Feature	What it does and why it is ahead	How to build it here	Gate
X10	Jamaica intelligence	JIC year-one and year-two tier switch with expiry warning; parish location factors for haulage; a resilience toggle that adds tie-downs and straps to the bill; GCT, 2 percent contractors levy and the 12.5 percent employer on-cost (already present) shown in one tidy statutory breakdown.	Rate book tiers (E4); extend the location-pack index to material and haulage separately (today the index scales materials only); resilience assembly in X7.	Every factor sourced and dated. Tier switch changes totals by exactly the book ratio.
X12	Genuinely offline and self-updating	Today the calculator page deliberately does not register the service worker, so a first-time or calculator-only visitor is not offline-capable and relies on a manual ?v bump. A scoped service worker with a versioned cache makes it installable, offline from cold start, and self-updating, and retires the stale-script trap.	A calculator-scoped worker that reuses the shell cache logic; rate-book updates arrive by importing JSON; update prompt on new version.	Cold offline start on a fresh device works. A version bump reaches an already-installed client without a hard refresh.
X13	Accessibility and language	Totals announced to screen readers, full keyboard use, WCAG 2.2 AA contrast in both themes, optional Spanish and French for the wider Caribbean market.	aria-live already exists on the result card; add checks to the Playwright audit harness (qa-calc-playwright-audit).	Audit harness passes at three viewports and both themes. Translated strings tested for overflow.

What I did not verify. The official JIC figures (I only have your extract); the second-year tier mechanics of the 19 percent increase; any carbon factor values; any exchange rate; hurricane-code requirements for tie-downs. Each is marked Verify or left to the rate book. No number in this plan should be copied into the product until it has a source and a date.

8. First actions

- Run npm run verify and both calculator harnesses on a clean copy and confirm green. This is the Phase 0 baseline.
- Get the official IMAJ rate sheet (876-929-8580, imaj.org.jm shop). It settles every Verify item and the open questions in G9 and G10.
- Decide the licensing route in section 4.
- Approve Phase 1 first. Fixing the currency behaviour (A1) matters regardless of any new trade, because JMD users are being shown relabelled USD rates today.

- Tell me which Verify, Added and Beyond items are in scope, so the phases are fixed before building starts.